

The DroneNav Story

How a Lifetime of Building Better Systems Led to the Future of Aviation

Working Manuscript

Chapter 1 — Before DroneNav

People often ask where an idea comes from.

Some believe great ideas arrive in a single flash of inspiration. My experience has been different. The ideas that have shaped my life did not appear overnight. They were built over decades through curiosity, mentorship, hard work, and a continual desire to understand how complex things work.

Long before there was a company called DroneNav, before I became a software architect, before I taught students how to build drones, I was simply a curious kid who loved to understand machines.

My father recognized that curiosity early. He wasn't content to answer my questions—he put tools in my hands. Together we worked on automobiles, engines, and electrical systems. His Army experience, along with the knowledge of his military friends, exposed me to electronics while I was still young. They didn't just teach me how to repair things; they taught me that every machine has a purpose and every problem has a solution if you are willing to understand how it works.

But my father gave me something even more valuable than technical skills.

He found mentors.

When my brother and I were young teenagers, my father introduced us to two retired Air Force pilots who owned a farm in a neighboring town. They operated a crop-dusting business from their private runway, but what fascinated me most wasn't watching the airplanes fly. It was discovering that they built many of those aircraft themselves.

For two summers, my mother packed our lunches every weekend, and we spent our days on that farm learning about aircraft construction. I remember wandering through their workshop, surrounded by aircraft parts, tools, and stacks of Air Force technical manuals. I was far too young to understand much of what those manuals contained, but I understood something much more important: airplanes were not magic. They were engineered systems built by people who understood the science behind them.

Those summers changed my life.

Unlike many people who become interested in aviation because they dream of becoming pilots, I found myself fascinated by something different. I wasn't drawn to flying an airplane as much as I

wanted to understand why it flew. I wanted to understand lift, propulsion, structures, and the engineering decisions that transformed thousands of individual parts into a machine capable of flight.

When those summers ended, I tried to build my own powered aircraft from parts I found in my father's garage.

I actually finished building it.

It never flew.

At the time, I was disappointed. Looking back, I realize that failure taught me an important lesson. Building something is only part of the challenge. Truly understanding the science behind it is what allows it to succeed.

That lesson would quietly follow me for the rest of my life.

Decades later, after building enterprise software systems, teaching engineering students, and mentoring young people in drone construction, I would often think back to those early experiences. I can still remember the smell of gasoline around that little airplane I tried so hard to build. Those memories have never faded.

Today, people know me as a software architect, inventor, educator, entrepreneur, and the founder of DroneNav. But none of those things happened by accident.

They began with a father who believed in mentorship, two retired Air Force pilots willing to invest their weekends in a curious teenager, and a young boy who wanted to understand not simply how to build a machine, but why it worked.

Looking back now, I realize they were teaching me something much bigger than aviation.

They were teaching me how to think.

Chapter 2 — Learning Discipline

By the time I reached my senior year of high school, two interests had firmly taken hold in my life: aviation and computers.

I had saved enough money to purchase my first computer, a Commodore PET 2001. For many people it was simply an early business computer. For me, it was a gateway into an entirely new

world. I taught myself BASIC and quickly became fascinated by the idea that software could instruct a machine to solve problems. It wasn't simply programming that interested me—it was the realization that computers could become tools for reasoning.

I entered college intending to study Computer Science. Academically, I wasn't intimidated by the material. The computer science courses came naturally. What I struggled with was something much more personal.

I lacked discipline.

Looking back, I simply wasn't mature enough. I had the curiosity and the technical ability, but not yet the focus required to turn those abilities into a profession.

Rather than continue down that path, I made a decision that would permanently shape my life.

I joined the United States Marine Corps Reserve.

The Marines gave me many things that I had been missing. They taught discipline, accountability, and responsibility. More importantly, they introduced me to aviation in a way I had never experienced before.

I trained as an Aviation Operations Specialist (MOS 7041).

For the first time, I wasn't looking at airplanes as machines to be built. I was looking at aviation as a complete operational system.

Every aircraft represented far more than a pilot and an engine. Behind every successful flight was an enormous amount of planning, coordination, documentation, communication, maintenance, logistics, and safety procedures. The Marine Corps demanded that every one of those activities be understood and executed correctly.

Unlike organizations with large technology budgets, the Marines still relied heavily on manual procedures. We learned to perform every task by hand because, in aviation, people must understand the process before they can trust the technology.

At first the workload seemed overwhelming.

Eventually it became something else.

It became a lesson.

As I watched highly trained professionals perform countless manual procedures simply to keep aircraft operating safely, a familiar thought began to return.

There has to be a better way.

That thought wasn't criticism.

It was curiosity.

The Marines taught me why those procedures existed. They also taught me something equally valuable: once you truly understand a complex system, you earn the right to improve it.

That lesson would become one of the defining principles of my career.

For the rest of my professional life, I would never begin by trying to replace a process. I would begin by understanding why the process existed in the first place.

Only then would I ask how technology could make it better.

Without realizing it, the Marines had given me a framework that would guide everything I would build in the years ahead—from enterprise software systems, to educational programs, and ultimately to DroneNav.

Chapter 3 — Reinvention

My career in software began at a small company developing airline revenue management systems.

Although the company itself was relatively small, the problems we solved were anything but.

Our customers were major airlines whose annual revenues measured in the hundreds of millions of dollars. Every night our software processed enormous quantities of data—forecasting passenger demand, optimizing seat inventory, and recommending pricing strategies that directly affected airline profitability the following morning.

Failure was not an option.

Every forecast had to be accurate.

Every optimization had to complete before sunrise.

Every minute of processing time mattered.

The software was written primarily in the C programming language because nothing else could deliver the performance required for the enormous mathematical models we were executing.

It worked remarkably well.

Until it didn't.

As the system grew over the years, every enhancement became increasingly risky. Business logic was tightly intertwined with the underlying software. A seemingly small change could ripple through the entire application, requiring weeks of testing and introducing the possibility of unintended consequences.

The system wasn't failing because the engineers lacked talent.

It was failing because we were asking software engineers to solve the wrong problem.

That realization took me back to something that had bothered me years earlier during my very first Computer Science course.

The professor referred to BASIC, FORTRAN, COBOL, and other programming languages as "languages."

Even then, something about that description never felt right.

Human languages exist so people can communicate ideas.

Programming languages existed primarily so people could describe instructions to machines.

Those are not the same problem.

For years, that thought quietly remained in the back of my mind.

Now, standing in the middle of one of the world's most sophisticated airline optimization systems, it suddenly made sense.

The people who truly understood airline revenue management were not software engineers.

They were business analysts.

They understood forecasting.

They understood pricing.

They understood optimization.

Yet every time they wanted to improve the system, they first had to explain their ideas to software engineers, who then translated those ideas into data structures and procedural code.

That translation process wasn't simply inefficient.

It introduced complexity that didn't need to exist.

Once again, a familiar thought appeared.

There has to be a better way.

What if the business expert could describe the problem directly?

What if software engineers stopped manually modeling every business concept?

What if the computer itself became responsible for constructing and managing the underlying model?

Those questions became the foundation of what would eventually become the Meta-Data Engine.

Instead of writing software that explicitly described every business rule, the Meta-Data Engine interpreted metadata supplied by domain experts and generated the structures necessary to execute complex forecasting and optimization processes.

It represented a fundamental shift in perspective.

Rather than teaching people to think like computers, I wanted computers to become better at understanding the intent of people.

Years later, many would describe this way of thinking as part of the evolution toward artificial intelligence.

At the time, I wasn't trying to invent AI.

I was simply trying to remove unnecessary barriers between human knowledge and computational power.

Looking back, I realize the Meta-Data Engine taught me one of the most important lessons of my career.

Technology achieves its greatest value not when it replaces people, but when it allows people to contribute their expertise more directly.

That lesson would eventually shape everything I built afterward.

Chapter 4 — Stewardship

By the time my professional career was well established, I had been fortunate to work on some of the most technically challenging software systems in the industry. Airline revenue management, healthcare, telecommunications, logistics, financial systems—each project presented new opportunities to solve complex problems and improve the way organizations operated.

Yet something else had begun to occupy my thoughts.

Teaching.

My wife had spent thirty years as an elementary school teacher.

She loved teaching.

She believed every child could learn when given the right environment, the right encouragement, and the right teacher.

As retirement approached, I realized something that surprised me.

I didn't want her to stop teaching.

Teaching wasn't simply her profession.

It was who she was.

That realization became the beginning of a new chapter in our lives.

I founded Pronouse Incorporated as the parent company for a number of ideas that combined consulting, education, science, and technology. One of those ideas became Progress Math.

Before opening the school, we spent considerable time studying educational research.

One conclusion appeared again and again.

Strong mathematical reasoning was one of the most consistent predictors of future success across nearly every profession.

To many people, mathematics is a collection of formulas.

To us, it represented something much deeper.

Mathematics teaches people how to reason.

It develops the executive function required to solve problems, recognize patterns, and make sound decisions.

Those are life skills.

Our goal was never simply to prepare children for better mathematics grades.

Our goal was to prepare them for better lives.

Over the years, students came to Progress Math as kindergartners and eventually returned years later to tell us how those early experiences had influenced the adults they became.

Those conversations remain some of the most rewarding moments of my life.

People sometimes ask whether I had a favorite student.

The answer has always been the same.

No.

Teachers cannot afford favorites.

The students most likely to succeed often need us the least.

Our responsibility is larger than that.

We have to reach as many children as possible.

We have to use our time wisely.

We have to develop systems, methods, and experiences that help every student grow.

In my mind, the measure of a teacher is not the success of one extraordinary student.

It is the number of lives quietly changed over time.

Looking back now, I realize that Progress Math taught me something that would later become essential to DroneNav.

Great systems are not built for exceptional people.

Great systems are built so ordinary people can achieve extraordinary results safely, confidently, and consistently.

Education and engineering had never really been separate pursuits.

Both were about building systems that help people succeed.

Chapter 5 — The Drones That Got Away

Teaching became one of the most rewarding chapters of my life.

Through Progress Math, Science ATL, Marietta High School, and the Technology Student Association, I had the privilege of introducing young people to engineering through drone construction. Many students had never held a soldering iron before arriving at camp. By the end of the day, they had assembled a functioning aircraft with their own hands.

Watching that first successful flight never became ordinary.

The excitement on a student's face when something they built actually lifted into the air reminded me why I loved teaching. Parents often stood nearby in disbelief. They weren't simply watching a drone fly. They were watching their children discover that they were capable of building something they had once believed impossible.

Those moments made every hour of preparation worthwhile.

But not every flight ended the way we hoped.

Occasionally, a drone would drift beyond the flying field.

Sometimes it was an inexperienced pilot.

Sometimes a flight controller malfunction.

Sometimes something we never fully understood.

Whatever the cause, the result was always the same.

The search began.

I spent countless hours walking through woods, neighborhoods, playgrounds, and vacant lots searching for aircraft that should never have left the flying area in the first place.

Every search carried the same concerns.

The drone itself represented weeks of preparation and significant expense.

More importantly, every aircraft carried a lithium polymer battery that could become dangerous if damaged.

I felt responsible for every one of them.

What surprised me most, however, was the reaction of the students.

Most of them weren't upset.

To them, the drone had already fulfilled its purpose.

They had built it.

They had flown it.

They had discovered something about themselves.

The experience remained with them even after the aircraft was gone.

I eventually realized that my students and I were solving two completely different problems.

They were learning.

I was trying to build a safer system.

After every camp I found myself asking the same question.

Why had we accepted this as normal?

Why had we accepted that drones simply fly away?

Why were we designing aircraft to recover after failure instead of designing a system that prevented failure in the first place?

At the time, I didn't have the answer.

I only knew one thing.

The problem wasn't going away.

Chapter 6 — The Decision

Looking back, people sometimes ask me when the idea for DroneNav first appeared.

The truth is, I don't think it began with an idea at all.

For years I had accepted runaway drones as one of the unfortunate realities of teaching.

My responsibility wasn't to invent new technology.

My responsibility was much simpler.

Keep the students safe.

Keep the public safe.

Recover every aircraft I possibly could.

That responsibility followed me through every camp.

Whenever a drone disappeared beyond the flying field, my first thoughts were never about replacing the aircraft.

I wondered where it had landed.

Was anyone nearby?

Was the lithium polymer battery damaged?

Could someone be hurt?

Only after those questions were answered did I think about the cost of replacing the drone.

Safety always came first.

I never imagined that those experiences would eventually lead me to build an entirely new aviation system.

Then one afternoon, everything changed.

A nonprofit organization asked me to host a drone camp for its summer program.

Unlike previous camps, where each family purchased its own drone kit, this organization would purchase every component for every student.

The investment represented thousands of dollars.

One afternoon I found myself alone in my workshop assembling those kits.

One by one, I laid out frames, motors, flight controllers, electronic speed controllers, propellers, batteries, and hardware for each student who would soon arrive.

As I worked, a thought quietly settled over me.

I already knew what was going to happen.

Some of these drones were going to fly away.

Not because the students would do anything wrong.

Not because the instructors would fail.

Not because the organization had made a poor investment.

They would fly away because I had accepted a system that made that outcome seem inevitable.

I stopped working.

For several minutes I simply looked across the room.

Every box represented a child who was about to discover engineering.

Every kit represented an organization that had trusted me with its resources.

Every aircraft represented a responsibility.

Until that afternoon, I had always focused on recovering drones after something went wrong.

For the first time, I asked a different question.

Why had I accepted that failure was unavoidable?

Why were we designing systems to respond after aircraft escaped rather than preventing them from escaping in the first place?

That was the moment everything changed.

I wasn't thinking about a software company.

I wasn't thinking about investors.

I wasn't thinking about patents.

I was thinking about responsibility.

I realized I had been solving the same safety problem over and over again.

Not because it couldn't be solved.

Because everyone—including me—had accepted it as normal.

That afternoon, I made a decision.

I would stop accepting it.

I didn't know what the solution would look like.

I didn't know how many years it would take.

I only knew one thing with certainty.

There had to be a better way.

That decision would eventually become DroneNav.

Chapter 7 — Thinking in Systems

By the time I made the decision to solve the problem of runaway drones, I wasn't thinking about software anymore.

I was thinking about transportation.

For most of my career I had worked on systems that managed complexity. Whether those systems forecast airline demand, optimized business operations, or taught children mathematics, they all shared one characteristic.

The best systems were rarely the most complicated.

They were the ones that removed unnecessary complexity.

As I began studying the future of low-altitude aviation, I noticed something that everyone else seemed willing to accept.

Nearly every conversation focused on making drones smarter.

Smarter navigation.

Smarter sensors.

Smarter obstacle avoidance.

Smarter autonomy.

Those were important advances.

But they all assumed the same thing.

That every aircraft should be responsible for solving the navigation problem on its own.

The more I thought about it, the less sense it made.

Imagine asking every automobile manufacturer to solve traffic management independently.

There would be no roads.

No intersections.

No speed limits.

No traffic signals.

No highway signs.

No shared rules.

Only increasingly intelligent automobiles trying to negotiate an increasingly chaotic environment.

We would never build a transportation system that way.

We built roads first.

The same is true for railroads.

Ships travel through designated channels.

Commercial aviation depends upon airports, controlled airspace, published procedures, and air traffic management.

Every successful transportation system shares something in common.

Infrastructure comes first.

Only then do vehicles operate safely within it.

That realization became the foundation of DroneNav.

I wasn't trying to build a smarter drone.

I was trying to build the equivalent of roads for the sky.

The first question became:

Where should drones be allowed to fly?

The second question followed naturally.

Where should they not fly?

From those simple questions emerged the first concepts that would eventually become DroneNav.

Communities needed designated operating areas where drone activity could begin and end.

I called those **Sites**.

They needed places where aircraft could safely launch, land, recharge, or exchange payloads.

Those became **DronePorts**.

They needed clearly defined pathways connecting one place to another.

Those became **Corridors**.

The word was intentional.

Internally I often used the word *route*, because engineers naturally think in terms of paths between two points.

But I eventually realized that *corridor* tells a much larger story.

A route belongs to one aircraft.

A corridor belongs to everyone.

A corridor is shared public infrastructure.

Communities also needed a way to identify areas requiring special consideration.

Schools.

Hospitals.

Parks.

Temporary events.

Sensitive facilities.

Some areas needed to prohibit flight entirely.

Others needed to define where flight must remain.

Those became ****Zones****.

As these ideas came together, another realization emerged.

Infrastructure alone would never be enough.

Roads exist because someone plans them.

Engineers design them.

Communities approve them.

Inspectors verify them.

Governments maintain them.

Every piece of transportation infrastructure exists because people trust the process that created it.

Why should drone infrastructure be any different?

That question fundamentally changed the architecture of DroneNav.

Every Site.

Every Corridor.

Every DronePort.

Every Zone.

Would follow a governance process before becoming operational.

Ideas would be proposed.

Surveyed.

Reviewed.

Approved.

Only then would they become part of the transportation network.

Many people think governance slows innovation.

I have come to believe the opposite.

Governance is what allows innovation to scale.

Without trust, infrastructure eventually becomes chaos.

Without accountability, safety eventually becomes chance.

Infrastructure and governance are not separate systems.

They are two halves of the same idea.

Looking back now, I realize DroneNav was never intended to be another drone application.

It was conceived as public infrastructure.

Not for one aircraft.

Not for one manufacturer.

Not even for one generation.

It was designed to become part of the foundation upon which the next fifty years of low-altitude aviation could safely be built.

Chapter 8 — Governance Before Autonomy

People often ask me whether autonomous aviation is inevitable.

I believe it is.

The more important question is whether it will be safe.

History has taught us that every transportation revolution follows a remarkably similar path.

The public remembers the vehicles.

History remembers the infrastructure.

The Romans are remembered for their roads.

The Industrial Revolution is remembered for its railroads.

The twentieth century transformed society through interstate highways, airports, navigation systems, and air traffic control.

Every generation inherits infrastructure that someone else had the wisdom to build.

Yet when I listen to conversations about the future of drones, I notice something curious.

Nearly every discussion begins with the aircraft.

Artificial intelligence.

Autonomous flight.

Computer vision.

Obstacle avoidance.

Machine learning.

These technologies are remarkable.

But they all answer the same question.

****How should the aircraft think?****

Very few people begin with a different question.

****What kind of airspace are we asking that aircraft to enter?****

To me, that is the more fundamental problem.

No amount of artificial intelligence can replace thoughtful infrastructure.

A brilliant aircraft operating within an unorganized environment remains part of an unorganized system.

Technology alone cannot create order.

Society does.

Roads create order.

Railroads create order.

Shipping lanes create order.

Airports create order.

The future of autonomous aviation deserves the same foundation.

That is why DroneNav begins with governance rather than autonomy.

Many people assume governance slows innovation.

I have reached the opposite conclusion.

Governance is what allows innovation to scale.

Without governance there is no shared trust.

Without shared trust there is no public acceptance.

Without public acceptance there is no transportation system.

There are only demonstrations.

This philosophy influenced every architectural decision within DroneNav.

Before an aircraft flies, communities should understand where flight is appropriate.

Before infrastructure becomes operational, experts should survey it.

Before the public depends upon it, independent reviewers should verify it.

Only then should autonomy begin.

Autonomy is not the first chapter of aviation.

It is the last.

It is something society earns after infrastructure, governance, and public confidence have been established.

That belief places DroneNav on a different path than many technology companies.

Our goal has never been to build the smartest drone.

Our goal is to build the infrastructure that allows every drone—regardless of manufacturer—to operate within a framework that communities understand and trust.

The future of aviation will not be determined solely by advances in artificial intelligence.

It will be determined by whether society has the wisdom to build the systems that allow intelligence to operate safely.

History will not judge autonomous aviation by the intelligence of its aircraft.

It will judge it by the wisdom of the infrastructure that surrounds them.

That is the future DroneNav is working to build.

Chapter 9 — Building for Generations

As engineers, we often think in terms of projects.

Requirements.

Schedules.

Deadlines.

Product releases.

But the systems that shape civilization are built differently.

They are built for people we will never meet.

The roads we travel today were designed by engineers who never knew our names.

The airports we use every day were planned decades before many of us were born.

The bridges we cross were built with the hope that future generations would trust them without ever thinking about the people who designed them.

That is the true purpose of infrastructure.

When it succeeds, it quietly becomes part of everyday life.

No one admires a stop sign.

No one celebrates a traffic signal.

No one thinks about the engineering behind an airport runway.

Yet society depends upon them every day.

My hope for DroneNav is much the same.

I do not hope it becomes famous.

I hope it becomes trusted.

I hope there comes a day when communities simply expect low-altitude aviation to operate safely because the infrastructure quietly does what it was designed to do.

If that day ever arrives, the greatest compliment DroneNav could receive is that people no longer think about it.

They simply trust it.

Over the course of my career I have learned that technology changes far more quickly than principles.

Programming languages evolve.

Computers become faster.

Artificial intelligence grows more capable.

Aircraft become more autonomous.

But certain ideas remain constant.

Understand the system before attempting to improve it.

Question unnecessary complexity.

Design for safety before convenience.

Teach what you have learned.

Leave every system better than you found it.

Those principles shaped the Meta-Data Engine.

They shaped Progress Math.

They shaped DroneNav.

Long after today's technology has been replaced, I hope those principles continue to guide the engineers who inherit whatever comes next.

When I was young, my father introduced me to people who changed my life.

At the time I believed they were teaching me how to build engines and airplanes.

Only much later did I understand what they had truly given me.

They had shown me that knowledge carries responsibility.

Every generation inherits systems built by those who came before.

Every generation has the opportunity—and the obligation—to improve those systems before passing them forward.

Looking back now, I realize that every chapter of my life has been another opportunity to honor that responsibility.

As a Marine, I inherited aviation procedures and learned why they mattered.

As a software engineer, I inherited complex systems and searched for better ways to express human knowledge.

As a teacher, I inherited students whose future depended upon the quality of the education we provided.

As the founder of DroneNav, I inherited an emerging form of aviation that had yet to receive the infrastructure it deserved.

Each chapter asked the same question.

How can I leave this better than I found it?

I do not know what low-altitude aviation will look like fifty years from now.

I hope it is safer.

I hope it is more accessible.

I hope it serves communities in ways we have not yet imagined.

Most of all, I hope future engineers continue asking the questions that have guided me throughout my life.

Are we doing something unnecessarily?

Are we solving the right problem?

Can we build something that serves everyone, not just ourselves?

If they do, they will create solutions far better than anything I could imagine today.

That is exactly what every generation should hope for.

When this story began, I was a curious boy trying to understand why an airplane flew.

Today I find myself asking much the same question in a different form.

How do we help an entire society fly safely together?

The tools have changed.

The responsibility has not.

If DroneNav contributes in some small way to answering that question, then I will consider my life's work a success.

The greatest systems are not remembered because they were revolutionary.

They are remembered because they quietly made the world a better place for those who followed.

That is the future I hope we leave behind.